

Algebra: Word Problems with Real Numbers



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Name: _____ Date: _____ Score: ____ / 8

DIRECTIONS

Translate each word problem into an algebraic expression or equation, then solve using integers, fractions, and decimals as needed.

1. Write as an integer: A diver is 15 meters below sea level. Answer: _____	2. Write as a fraction and decimal: 3 out of 8 students passed the quiz. Answer: _____
3. Translate then simplify: The sum of -4 and 9, decreased by 3. Answer: _____	4. Find the additive inverse: The additive inverse of -7. Answer: _____
5. Solve: Maria owed \$45. She paid \$18 back. How much still owed? Answer: _____	6. Solve: A tank lost $\frac{1}{4}$ of its 80 L. How much water remains? Answer: _____
7. Solve: Temp dropped 3.5° each hour for 4 hours. Total change? Answer: _____	8. Solve: Add -12, 8, and the additive inverse of -5. Answer: _____

Based on the Numberbender lesson "Word Problems with Real Numbers" • youtu.be/Ktp95pPYUdw

Page 1 of 2 — Worksheet



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Answer Key & Solutions

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TEACHER NOTES

Items 1-2: translating real-world situations into integers. Items 3-4: additive inverse. Items 5-6: fraction and decimal word problems. Items 7-8: multi-step real-number problems. Encourage students to define a variable before writing the equation.

1. Write as an integer: A diver is 15 meters below sea level.

Answer: -15

Below sea level = negative; 15 m below = -15.

2. Write as a fraction and decimal: 3 out of 8 students passed the quiz.

Answer: $3/8 = 0.375$

$3 \div 8 = 0.375$.

3. Translate then simplify: The sum of -4 and 9, decreased by 3.

Answer: 2

$(-4 + 9) - 3 = 5 - 3 = 2$.

4. Find the additive inverse: The additive inverse of -7.

Answer: 7

Additive inverse of -7 is 7 (they sum to 0).

5. Solve: Maria owed \$45. She paid \$18 back. How much still owed?

Answer: -\$27

$-45 + 18 = -27$; she still owes \$27.

6. Solve: A tank lost $1/4$ of its 80 L. How much water remains?

Answer: 60 L

$1/4$ of 80 = 20 removed; $80 - 20 = 60$ L.

7. Solve: Temp dropped 3.5° each hour for 4 hours. Total change?

Answer: -14°

$-3.5 \times 4 = -14^\circ$ total drop.

8. Solve: Add -12, 8, and the additive inverse of -5.

Answer: 1

Additive inverse of -5 is +5; $-12 + 8 + 5 = 1$.

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Page 2 of 2 — Answer Key

